Evaluating Uncertainty Estimation

How to evaluate the quality of uncertainty estimates?

- Proper Scoring Rule (LL, Brier Score)
  - [Gneiting and Raftery, “Strictly Proper Scoring Rules, Prediction, and Estimation”, JASA 2007]
- Calibration (NLL, ECE)
  - [“Obtaining Well Calibrated Probabilities Using Bayesian Binning”, AAAI 2015]
- Out-of-distribution detection



Evaluating Uncertainty Estimation: Out-of-distribution detection

OoD: Motivation

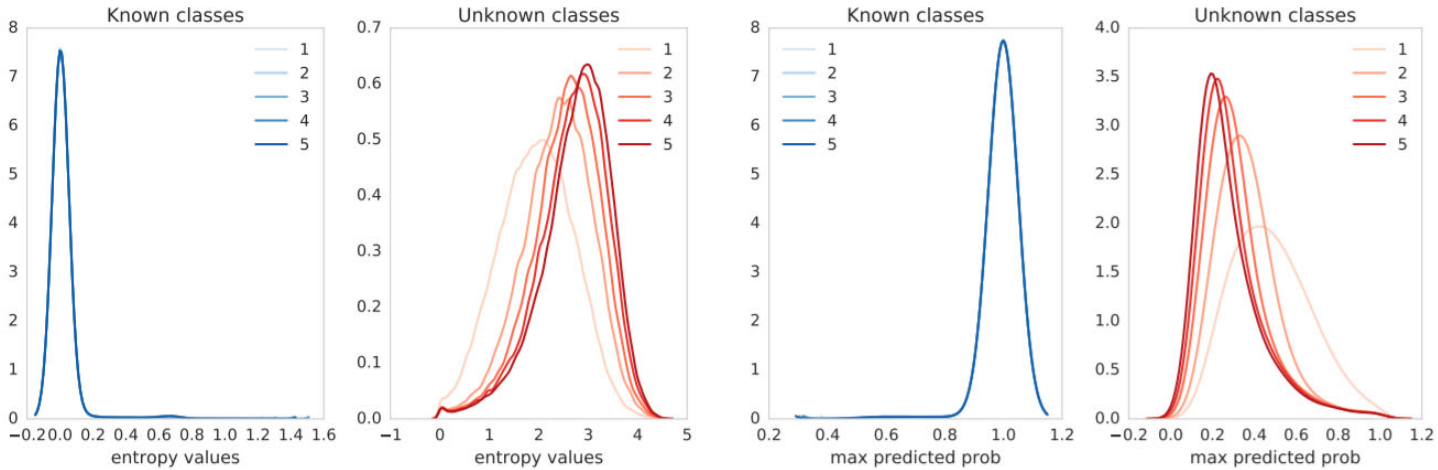
- In machine learning, we usually assume the test data comes from the same distribution as the training data.
- This is frequently violated in practice.
- What should the model do in those situations?
- One possible solution is to identify test-time out-of-distribution samples and abstain from prediction
- As such, this can be an important benchmark for uncertainty estimation
  - Out-of-distribution samples should induce high epistemic (distributional) uncertainty.



Evaluating Uncertainty Estimation: Out-of-distribution detection

OoD: Example

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  - Out-of-distribution samples should induce high epistemic (distributional) uncertainty.
  - Example: ensemble of deep networks trained on ImageNet's ~200 dog classes (known) and tested on non-dog classes (unknown)



Evaluating Uncertainty Estimation: Out-of-distribution detection

OoD: Pitfalls of Uncertainty Estimation

- ["Benchmarking Neural Network Robustness to Common Corruptions and Perturbations", ICLR 2019]
- ["Can You Trust Your Model's Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift", NeurIPS 2019]
- ["Pitfalls of In-Domain Uncertainty Estimation and Ensembling in Deep Learning", ICLR 2020]



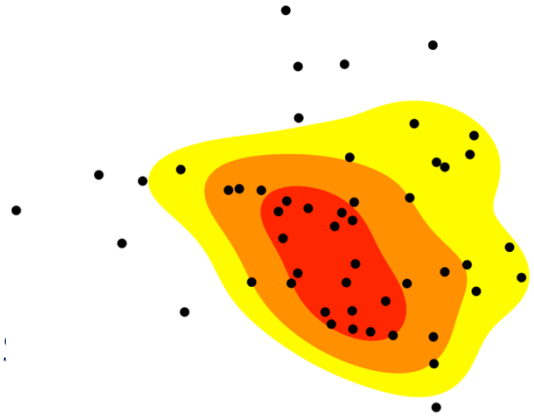
Evaluating Uncertainty Estimation: Out-of-distribution detection

OoD: Explicit Techniques

- OoD Detection has extensive application relevance
- There are methods designed to directly tackle it as a “classification” problem in both supervised and unsupervised way
- Examples
  - [“A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks”, ICLR 2017]
  - [“Deep Anomaly Detection with Outlier Exposure”, ICLR 2019]
  - [“Hyperparameter-Free Out-of-Distribution Detection Using Cosine Similarity”, ACCV 2020]
  - [“Generalized ODIN: Detecting Out-of-distribution Image without Learning from Out-of-distribution Data”, CVPR 2020]

Evaluating Uncertainty Estimation: Out-of-distribution detection

OoD: Density Estimation



- Some successful approaches
  - "A Simple Unified Framework for Detecting Out-of-Distribution Attacks", NeurIPS 2018
  - "Detecting Out-of-Distribution Examples with Gram Matrices", ICML 2020
- General deep generative models are not best method at the moment though
  - ["Do Deep Generative Models Know What They Don't Know?", ICLR 2019]
  - ["Understanding Failures in Out-of-Distribution Detection with Deep Generative Models", ICML 2021]

Next topic (Generative Models)!